

Multiple Choice (10 marks)

1. A meter stick with a speed of $0.8c$ moves past an observer. In the observer's reference frame, how long does it take the stick to pass the observer ?
 - (a) 1.6 ns
 - (b) 2.5 ns
 - (c) 4.2 ns
 - (d) 6.9 ns
2. A grating spectrometer can just barely resolve two wavelengths of 500 nm and 502 nm, respectively. Which of the following gives the resolving power of the spectrometer?
 - (a) 2
 - (b) 250
 - (c) 5,000
 - (d) 10,000
3. The sign of the charge carriers in a doped semiconductor can be deduced by measuring which of the following properties?
 - (a) Specific heat
 - (b) Thermal conductivity
 - (c) Electrical resistivity
 - (d) Hall coefficient
4. The speed of light inside of a nonmagnetic dielectric material with a dielectric constant of 4.0 is
 - (a) 1.2×10^9 m/s
 - (b) 3.0×10^8 m/s
 - (c) 1.5×10^8 m/s
 - (d) 1.0×10^8 m/s
5. The negative muon, μ^- , has properties most similar to which of the following?
 6. Electron
 7. Meson
 8. Photon
 9. Boson

True / False (10 marks)

Answer True or False

6. True or False: Electromagnetic radiation emitted from a nucleus is most likely in the form of gamma rays.
7. True or False: The primary source of the Sun's energy is the fusion of four hydrogen atoms into one helium atom, releasing energy according to $E=mc^2$.
8. True or False: The resolving power of the grating spectrometer for wavelengths of 500 nm and 502 nm is 250.
9. True or False: A dye laser is the best type of laser for spectroscopy across a range of visible wavelengths.
10. True or False: In a single-electron atom with $l = 2$, there are only 3 allowed values for the magnetic quantum number m_l .

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the allowed transitions for an electron in the $n = 4$, $l = 1$ state of hydrogen. Explain which final states are not reachable and why, using quantum mechanical principles.
12. Analyze a one-dimensional inelastic collision where a particle of mass $2m$ collides with a stationary mass m . Discuss the initial kinetic energy and calculate the fraction lost during the collision.

End of Paper